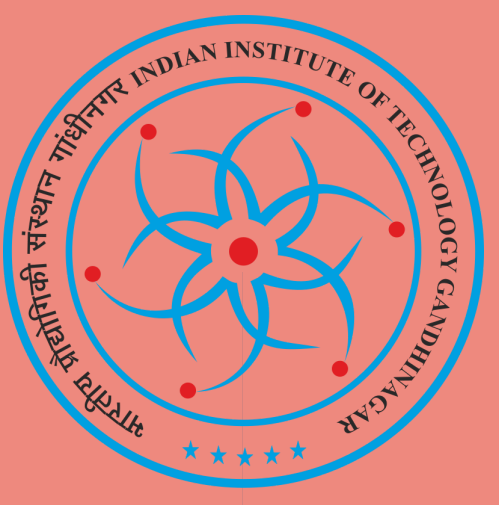




DESIGN OF PORTABLE UNIVERSAL TESTING MACHINE FOR TESTING MATERIALS AND STRUCTURAL MODELS

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INTRODUCTION

Material testing is vital in engineering to evaluate strength, stiffness, and ductility. Universal Testing Machines (UTMs) are commonly used for this purpose, but conventional UTMs are bulky, expensive, and non-portable, limiting their use in classrooms and field setups.

This project focuses on designing a compact, affordable UTM capable of applying up to 1 kN of load. The goal is to create a lightweight, modular, and user-friendly machine for testing small-scale models like academic specimens and bridge prototypes, without compromising accuracy or safety.

DESIGN PROCESS

The design of the portable Universal Testing Machine (UTM) involved several key approaches. The frame was constructed using lightweight aluminium extrusions of dimensions 30×30 mm with four 800mm high pillars, which withstand the reaction force during testing for keeping the structure stable, and a base with a steel plate and 20×20 mm aluminium grid structure providing a flat surface. This grid structure protects the base from wearing out by thrust bearings. The Frame also contain two 20 mm ball screws driven by NEMA 23 stepper motors, and a 640×40×50 mm aluminium movable bar integrated with an S-type load cell and load applicator. The machine operates using a displacement-controlled approach, where an Arduino Uno is programmed to regulate a predefined displacement rate (such as 2 mm/s). These signal drives the stepper motor to convert rotational input in lead screw into linear vertical motion in load applier for smoothly applying a load on specimen. The readings will be shown on a connected screen, which displays.

The simulations are carried out in Autodesk Fusion 360 to verify the structural safety and performance by analysing stress distribution, deformation, and safety factors under a 1 kN load

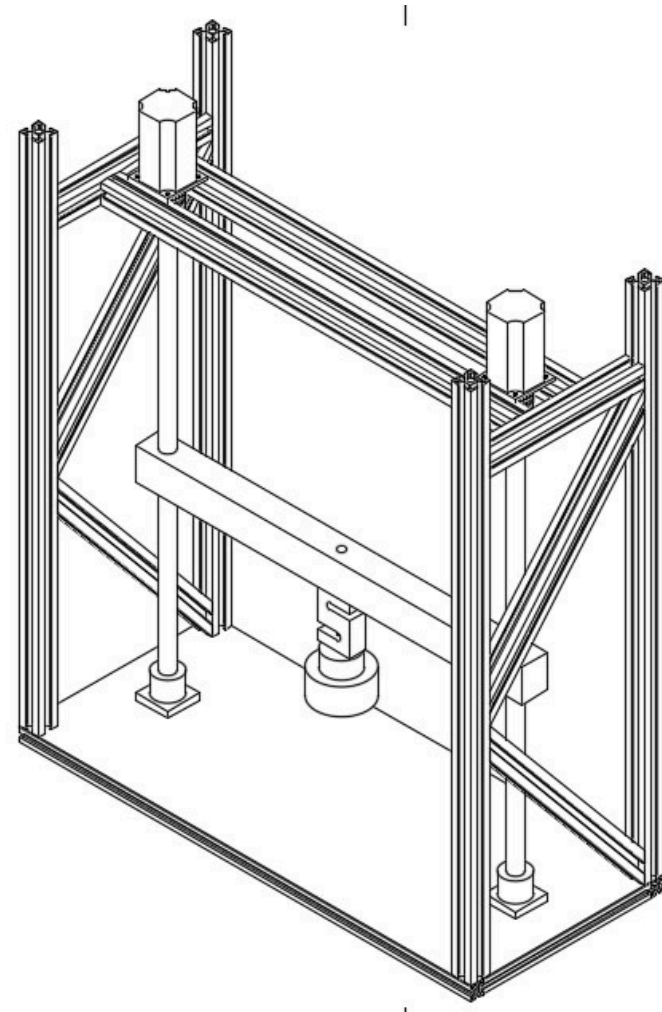


Fig 1: Final Design Of UTM

RESULT AND DISCUSSION

A separate structural analysis was conducted for the frame and the lead screw mechanism of the Universal Testing Machine (UTM), as we encountered challenges in simulating the entire assembly together within Fusion 360. From the analysis

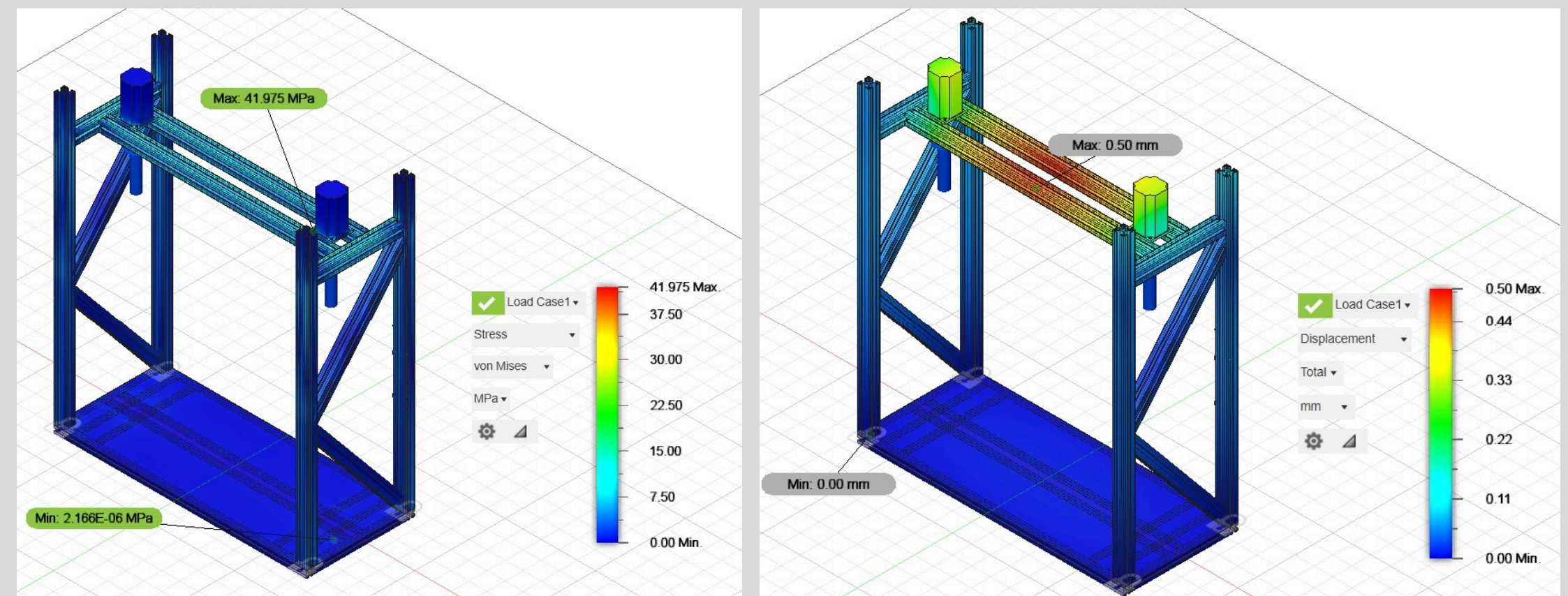


Fig 2 & 3 : Simulation Result for Frame

Finite Element Analysis (FEA) was conducted on the final CAD model by applying a total load of 500 N on each stepper motor.

The analysis showed a maximum von Mises stress of 42 MPa, which is well below the 240 MPa yield strength of aluminum 6061-T6 – the material used for the frame – confirming structural safety.

The maximum deflection, measured at 0.50 mm, occurred at the center of the horizontal crosshead. This is due to slight bending from the force applied symmetrically through both lead screws.

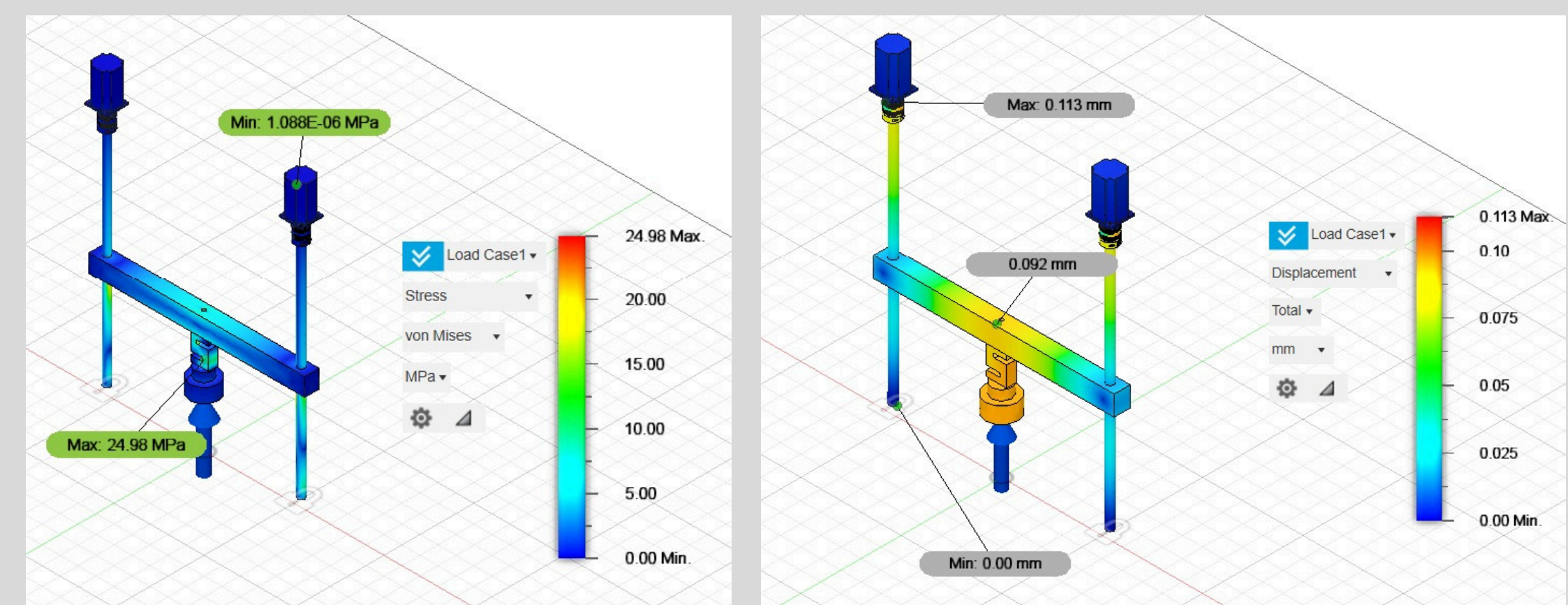


Fig 4 & 5 : Simulation result for lead screw mechanism connected with movable crosshead

Finite Element Analysis (FEA) was conducted on the assembly of lead screw mechanism and the load applier mechanism by applying a total load of 1000 N, on the load applier.

When a 1,000 N load was applied through the load applier, the maximum von Mises stress observed was 25 MPa, well below the yield strength of aluminum and mild steel, confirming the design's safety.

The maximum deflection recorded was 0.113 mm, occurring near the shaft coupler. This minimal deformation indicates high stiffness and ensures reliable performance during actual loading conditions

FUTURE WORK

We are now moving on to the fabrication stage of our portable Universal Testing Machine (UTM), where the virtual model will be turned into a working prototype, after finishing the design phase. The choice of material is being made with consideration for weight efficiency, strength, and cost. The UTM measures applied force using an S-type load cell in a displacement-controlled manner. To preserve portability and simplify the design, an LVDT sensor for direct displacement output was not included, despite its consideration. Following the acquisition of every component, fabrication and assembly will start.

CONCLUSION

For academic and small-scale material testing, our portable Universal Testing Machine (UTM) is designed to be lightweight, affordable, and dependable. Up to 1 kN of load may be securely supported by the design, which was created and verified using thorough 3D CAD modeling and FEA simulations in Autodesk Fusion 360. The stability of the structure with stresses well within material limits was validated by simulations.

The UTM is not industrial-grade, but its mobility and ease of use make it ideal for prototype testing and educational labs. After finishing the design stage, we are moving on to the fabrication stage to create a functional prototype that will enable us to use our virtual model in the real world.

[1] J. MATHEW and V. Francis, "Development, Validation and Implementation of Universal Testing Machine," DIVA, 2020. <https://www.diva-portal.org/smash/record.jsf?dswid=5778&pid=diva2%3A1472019> (accessed Jul. 03, 2025).

[2] M. Singh, "Development of a portable Universal Testing Machine (UTM) compatible with 3D laser-confocal microscope for thin materials," Advances in Industrial and Manufacturing Engineering, vol. 4, p. 100069, May 2022, doi: <https://doi.org/10.1016/j.aime.2022.100069>.